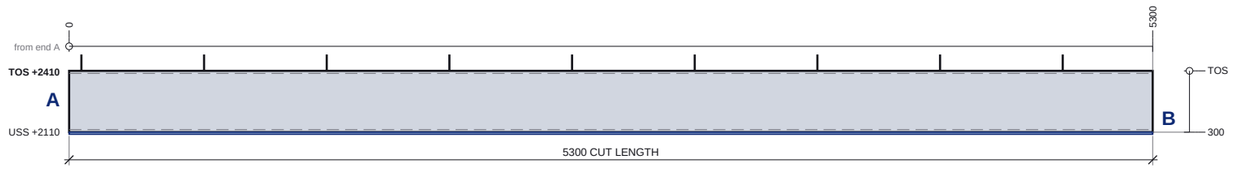
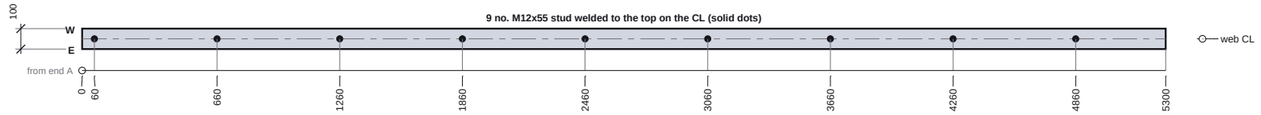
SB4 ELEVATION

Scale 1:30 | viewed from the E side, end A (S) left



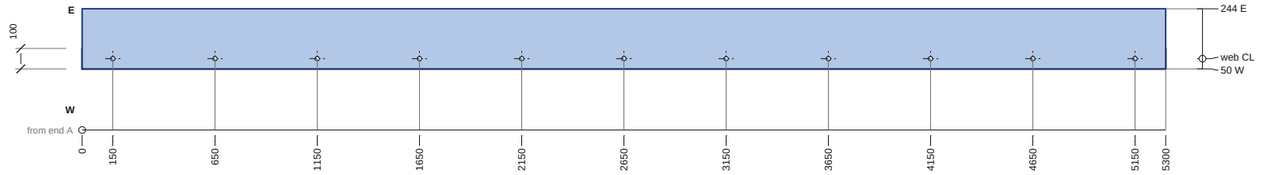
SB4 TOP - PLAN

1:30



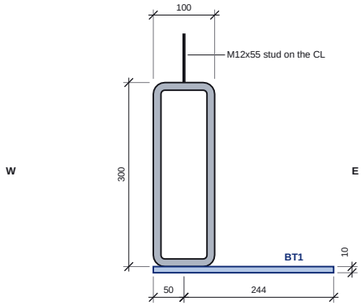
SB4 UNDERSIDE - VIEWED FROM BELOW

1:30



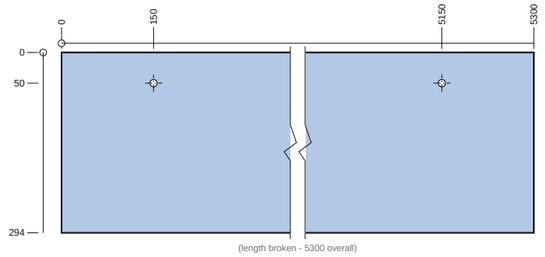
SECTION THROUGH SB4 1:10

looking from end A to end B: W left, E right



BT1: 50 W / 244 E of the CL, underside 310 below TOS

BT1 PLATE DETAIL 1:10



u along the beam from the south end (left edge), v from the west edge (top edge)

5300 x 294 x 10 thk S275
11 no. Ø12 holes
122.22 kg qty 1 on SB4
Hole u (all): 150, 650, 1150, 1650, 2150, 2650, 3150, 3650, 4150, 4650, 5150 (middle ones in the broken length)
Weld: 6mm FW both toes to the RHS (flare-bevel at the RHS corners, throat 5 min); ends 6 FW to the post inner faces
boot plate under the RHS, 10 back from the stone face; Ø12 vents in the 100 lap under the RHS (sealed lap, BS EN ISO 14713-2), plugged after galvanising - galvaniser to confirm
TLS detail - approved by the engineer

HOLE SCHEDULE

No.	FACE	X FROM A	POSITION	HOLE / STUD	PURPOSE
S1	top	60	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - approved by the engineer]
S2	top	660	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - approved by the engineer]
S3	top	1260	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - approved by the engineer]
S4	top	1860	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - approved by the engineer]
S5	top	2460	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - approved by the engineer]
S6	top	3060	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - approved by the engineer]
S7	top	3660	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - approved by the engineer]
S8	top	4260	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - approved by the engineer]
S9	top	4860	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - approved by the engineer]
BT1 V1..V11	BT1	150..5150 @500	u / v: BT1 PLATE DETAIL	Ø12	11 no. vent / drain holes in BT1 - plug / seal after galvanising

CUT LIST

MARK	ITEM	GRADE	LENGTH	QTY	KG
SB4	RHS300x100x12.5	S355J2H	5300	1	383.2
BT1	PL 5300x294x10	S275		1	122.22
TOTAL					**505.4 kg**

CONNECTIONS

Detail GP1 knee: SB4 end A/B on PC1/PC2 - Shop-welded knee (R4): RHS square-cut ends welded all round to the post inner faces, 8 FW; BT1 ends 6 FW to the post faces. Frame galvanised as one piece (galvaniser to confirm the bath + dip; temporary bolted spreader between the bases until set; post centres +/-3 after galvanising). Approved by the engineer 13.10.26.

NOTES

1. **APPROVED FOR CONSTRUCTION - the engineer approved the steelwork and the TLS details on 13.10.26; fabricate to these drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. BS EN 1090-2 EXC2, CE marked. RHS S355J2H (EN 10210-1), plates S275JR.
4. Holes: normal clearance (EN 1090-2 Table 11): M12 = 14, M16 = 18. Drilled or punched full size, deburred.
5. Running dimensions from end A / base underside (o = datum). Levels above FFL (model 0).
6. Coating: Hot-dip galvanised to BS EN ISO 1461 (thickness to Table 3); T-wash + paint where exposed (architect).
7. Holes / plates marked TLS detail are not on the engineer's drawings: approved by the engineer.
8. GP1 - TLS design R4, approved by the engineer 13.10.26 (replaces S-0010 UB203x133x30); calc on the GP1 sheet
9. RHS side walls land on the SHS corner radius (flat +/-45): flare-bevel weld, throat 5 min (approved by the engineer).
10. Studs welded before galvanising: chase the threads / oversize-tapped HDG nuts (BS EN ISO 10684).

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue
C04	13.10.26	Approved by the engineer 13.10.26 - issued for construction

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

SB4 RHS300X100X12.5 FABRICATION

GP1 portal beam between the posts over the 5300 garage door; carries GT1 + ~0.5 m stone on the boot

SCALE	SIZE	DATE	DRAWN
1:30 / details as noted	A3	13.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-ST-SB4	C04

Do not scale. All dimensions in mm. Check all dimensions on site.